



September 2018 Examination

EC400

Probability and Statistical Inference - SOLUTIONS

(SOLUTIONS ALLOCATED ONLY 92 MARKS): FINAL GRADE IS RESCALED TO 100!

Instructions to candidates

Time allowed: 2 hours

This paper contains **SEVEN** questions. Answer **ALL** questions.

On your script, write

- Your name
- Your class group number
- The name of your class teacher.

You are not allowed to use an electronic calculator

1. **(7 points - 10 minutes)** Let $f(x) = (x + 1)/2$, $-1 < x < 1$, zero otherwise. Find

(a) **3 points** The question was intended have stated $-1 \leq x \leq 1$, in which case the mode is 1 as intended (point with highest probability density). Because the question stated used the open interval,unintentionally, the answer should have been that the mode didn't exist. Both answers accepted.

(b) **2 points** The 25th percentile p is defined by the relation $\int_{-\infty}^p f(x)dx = 0.25$

$$\int_{-1}^p (x + 1)/2 dx = \frac{1}{4}x^2 + \frac{1}{2}x \Big|_{-1}^p = \frac{1}{4}p^2 + \frac{1}{2}p - \left(\frac{1}{4} - \frac{1}{2}\right)$$

1 point Therefore p needs to satisfy: $\frac{1}{4}p^2 + \frac{1}{2}p = 0$ or $p(\frac{1}{4}p + \frac{1}{2}) = 0$ meaning $p = 0$ or $p = -2$.

1 point The latter being ruled out by the domain of X , so $p = 0$.

2. **(8 points - 10 minutes)** If X has the following exponential distribution $f(x) = \lambda e^{-\lambda x}$, $0 < x < \infty$, find the probability density function of Y where $Y = \ln(\mu + \sigma X)$.

We recognize that $y = \ln(\mu + \sigma x)$ is a one-to-one transformation, so we can define its inverse: $x = \frac{e^y - \mu}{\sigma} \equiv v(y)$ with $v'(y) = \frac{e^y}{\sigma}$ We also observe that $y > \ln(\mu)$

$$\begin{aligned} g(y) &= f(v(y)) \cdot |v'(y)| \\ &= \begin{cases} \lambda e^{-\lambda \left(\frac{e^y - \mu}{\sigma}\right)} \cdot \frac{e^y}{\sigma} & \text{for } y > \ln(\mu) \\ 0 & \text{otherwise} \end{cases} \end{aligned}$$

3 points: use of proper formula to obtain marginal of Y

2 points: proper execution of formula

1 point: statement that formula uses the one-to-one relation between the r.v.'s

2 points: to properly state $y > \ln(\mu)$

3. **(12 points - 15 minutes)** Let X is a random variable with density

$$f(x) = \theta x^{\theta-1}, 0 < x < 1, 0 < \theta < \infty, \text{ zero elsewhere.}$$

(a) **4 points** $E(X) = \int_0^1 x f(x) dx = \int_0^1 \theta x^\theta dx = \left. \frac{\theta}{\theta+1} x^{\theta+1} \right|_0^1 = \frac{\theta}{\theta+1}$

(b) **4 points** The MME chooses $\hat{\theta}$ to ensure that its sample analogue is satisfied:

$$\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i = \frac{\hat{\theta}_{MM}}{\hat{\theta}_{MM} + 1}$$

upon rewriting this means:

$$\hat{\theta}_{MM} = \frac{\bar{X}}{1 - \bar{X}}$$

(c) **2 points** For consistency we want to show $\text{plim } \hat{\theta}_{MM} = \theta$.

Using Slutsky we have

$$\text{plim } \hat{\theta}_{MM} = \frac{\text{plim } \bar{X}}{1 - \text{plim } \bar{X}}$$

as $\text{plim } \bar{X} = E(X) = \frac{\theta}{\theta+1}$, we get

$$\text{plim } \hat{\theta}_{MM} = \frac{\frac{\theta}{\theta+1}}{1 - \frac{\theta}{\theta+1}} = \frac{\frac{\theta}{\theta+1}}{\frac{1}{\theta+1}} = \theta$$

consistency is proved.

2 points For unbiasedness we would like to argue $E(\hat{\theta}_{MM}) = \theta$. As the above suggests, that would be proven if we could bring the expectation operator in (just as we did with the plim operator). Unfortunately Jensen's inequality tells us that we cannot do this. So in general, our MM estimator will be biased.

4. **(13 points - 15 minutes)** Let (X, Y) have a joint density function

$$f_{X,Y}(x, y) = x + y, \quad 0 \leq x \leq 1, \quad 0 \leq y \leq 1, \quad \text{zero elsewhere.}$$

$$f_X(x) = \int_0^1 (x + y) dy = xy + \frac{1}{2}y^2 \Big|_0^1 = x + \frac{1}{2}, \quad 0 \leq x \leq 1, \quad \text{zero elsewhere}$$

$$f_{Y|X}(y|x) = \frac{f_{X,Y}(y,x)}{f_X(x)} = \frac{x+y}{x+1/2} \quad \text{with } 0 \leq x \leq 1, \quad 0 \leq y \leq 1, \quad \text{zero elsewhere}$$

$$E(Y|X=x) = \int_{-\infty}^{\infty} y f_{Y|X}(y|x) dy = \int_0^1 y \frac{x+y}{x+1/2} dy = \frac{1}{x+1/2} \int_0^1 (yx + y^2) dy = \frac{1}{x+1/2} \left[\frac{1}{2}y^2x + \frac{1}{3}y^3 \right]_0^1 = \frac{\frac{1}{2}x + \frac{1}{3}}{x+1/2}, \quad 0 \leq x \leq 1$$

By law of iterated expectations: $E(Y) = E_X[E(Y|X=x)]$

Student could derive the unconditional expectation either way (i) $E_X[E(Y|X=x)] = \int_0^1 \frac{\frac{1}{2}x + \frac{1}{3}}{x+1/2} f_X(x) dx = \int_0^1 \left(\frac{1}{2}x + \frac{1}{3} \right) dx = \left[\frac{1}{4}x^2 + \frac{1}{3}x \right]_0^1 = \frac{1}{4} + \frac{1}{3} = \frac{7}{12}$ and (ii) $E(Y) = \int_{-\infty}^{\infty} y f_Y(y) dy = \int_0^1 y \left(y + \frac{1}{2} \right) dy = \left[\frac{1}{3}y^3 + \frac{1}{4}y^2 \right]_0^1 = \frac{7}{12}$

4 points: proper definitions of conditional and unconditional expectations

3 points: *proper execution to obtain conditional expectation*

3 points: *proper execution to obtain unconditional expectation*

3 points: *clear discussion of law of iterated expectations*

5. **(12 points - 15 minutes)** The moment generating function (MGF), when it exists, completely determines the distribution of a random variable. Let $M(t)$ be the moment generating function of the continuous random variable X .

(a) **4 points** $M(t) = E(e^{tX}) = \int_{-\infty}^{\infty} e^{tx} f(x) dx$

(b) **2 point** With $R(t) = \ln(M(t))$, observe $R'(t) = \frac{M'(t)}{M(t)}$ and, using the quotient rule of differentiation, $R''(t) = \frac{M''(t)M(t) - M'(t)M'(t)}{M(t)^2}$.

4 points Pointing out that for the MFG: $M'(0) = E(X)$ and $M''(0) = E(X^2)$

1 points Observing that $R'(0) = \frac{1}{M(0)} M'(0) = M'(0) = \mu$

1 points Observing that $R''(0) = \frac{M''(0)M(0) - M'(0)M'(0)}{M(0)^2} = M''(0) - (M'(0))^2 = E(X^2) - [E(X)]^2 \equiv \text{Var}(X) = \sigma^2$

6. **(25 points - 30 minutes)** Let X equal the forced vital capacity (FVC) in liters for a female college student. (This is the amount of air that a student can force out of her lungs.). Assume that the distribution of X is (approximately) $N(\mu, \sigma^2)$ and σ^2 is unknown. A volleyball coach claims that the FVC of volleyball players is greater than 3.4 liters. She plans to test her claim using a random sample of size $n = 9$.

(a)

$$H_0 : \mu = 3.4, H_1 : \mu > 3.4$$

3 points

(b)

$$T = \frac{\bar{X} - 3.4}{SE(\bar{X})} = \frac{\bar{X} - 3.4}{\sqrt{S_x^2/n}} \sim t_{n-1} \text{ under } H_0 \text{ here } n = 9$$

3 points: proper test statistic

3 points: distribution under the null

0 points for Z statistic

(c) For the sample, she obtains $\bar{x} = 3.7$ and $s^2 = 1/9$.

- i. **4 points:** We reject if $t > t_{5\%,8}$ where $\Pr(T > t_{5\%,8}) = 5\%$, i.e., $t_{5\%,8} = 1.86$. Given the realisation, she will obtain $t = \frac{3.7-3.4}{\sqrt{1/9^2}} = 2.7$. Since 2.7 exceeds the critical value, she will reject the H_0 .

1 point: She found evidence in favour of her claim that volleyball players have a higher FVC.

- ii. The p-value gives the lowest level of significance at which you want to reject the null.

$$\Pr(T > 2.7)$$

2 points: proper statement on how you obtain the p-value

3 point: graph depicting the p-value

- iii. Here we need to obtain the probability of rejecting the null when 3.8 is the true mean:

$$\begin{aligned} & \Pr\left(\frac{\bar{X} - 3.4}{SE(\bar{X})} > 1.86 \mid \mu = 3.8\right) \\ &= \Pr\left(\frac{\bar{X} - 3.8}{SE(\bar{X})} > 1.86 - \frac{0.4}{SE(\bar{X})}\right) \\ &= \Pr(T_8 > 1.86 - \frac{0.4}{\sqrt{1/9^2}}) = \Pr(T_8 > -1.74) \end{aligned}$$

3 points: proper statement on how to obtain the power

2 points: proper restandardization to account for alternative (no need to see number -1.74)

1 point: refer to statistical table of student t distribution with 8 df for answer

7. **(15 points - 18 minutes)** Let $X_1, X_2, \dots, X_n$ be a random sample of size n from an exponential distribution with unknown mean equal to θ . Using Moment Generating Functions it can be shown that the random variable $W = \frac{2}{\theta} \sum_{i=1}^n X_i$ is $\chi^2(2n)$. Let $n = 7$ (erroneously stated $n = 9$ on paper, but corrected)

- (a) **(8 points)** To obtain the confidence interval we need to make use of

$$P(6.56 \leq W \leq 23.68) = 0.90$$

(provided in question). Next we rewrite the inequality to allow us to make a statement such that $P(a \leq \theta \leq b) = 0.90$ in order for $[a, b]$ to yield the required

90% confidence interval.

$$P\left(6.56 \leq \frac{2}{\theta} \sum_{i=1}^9 X_i \leq 23.68\right) = 0.90 \text{ or}$$

$$P\left(6.56 \leq \frac{2 * 9\bar{X}}{\theta} \leq 23.68\right) = 0.90$$

Take reciprocal (invert inequalities):

$$P\left(\frac{1}{6.56} \geq \frac{\theta}{18\bar{X}} \geq \frac{1}{23.68}\right) = 0.90$$

Multiply by $18\bar{X}$ (positive) :

$$P\left(\frac{18\bar{X}}{6.56} \geq \theta \geq \frac{18\bar{X}}{23.68}\right) = 0.90$$

Given our realisation for $\bar{X}$, our 90% confidence interval is given by $\left[\frac{18 \times 23.68}{23.68}, \frac{18 \times 93.6}{6.56}\right]$.

2 points: Definition of 90% confidence interval

2 point: Getting to the stage: $P\left(6.56 \leq \frac{2 * 9\bar{X}}{\theta} \leq 23.68\right) = 0.90$ (n instead of 9 fine)

3 points: Rearranging the inequalities to yield $P\left(\frac{18\bar{X}}{6.56} \geq \theta \geq \frac{18\bar{X}}{23.68}\right) = 0.90$

1 point: $\left[\frac{18 \times 23.68}{23.68}, \frac{18 \times 93.6}{6.56}\right]$

- (b) **(7 points)** The interval is useful when conducting a two sided hypothesis of θ with a 10% level of significance. Say $H_0 : \theta = \theta_0$, $H_1 : \theta \neq \theta_0$, and we should reject at the 10% level of significance if θ_0 does not lie in the 90% confidence interval.

2 points: Reference to the use of a two sided test, i.e. $H_0 : \theta = \theta_0$, $H_1 : \theta \neq \theta_0$

2 points: Use of 10% level of significance for our hypothesis test

3 points: Statement that relates θ_0 lying in the 90% confidence interval as our decision not to reject the null at the 10% level of significance and θ_0 not lying in the 90% confidence interval as our decision to reject the null at the 10% level of significance